

BPS61 – Lab 3: Architecture Selection & Written Justification

System: Household Carbon Footprint & Target Tracker

Selected Architecture: Layered Architecture

Architectural Choice

We chose **Layered Architecture** for the Household Carbon Footprint & Target Tracker. The system is separated into a Presentation Layer for resident interaction, a Business Layer for consumption processing, CO₂e calculation, footprint monitoring and target management, and a Data Layer for persistent household carbon-footprint information.

Reason 1 – Clear Separation of Responsibilities

The BPS61 requirements cover distinct operations: recording monthly electricity and transport data, calculating monthly CO₂e, viewing monthly and historical footprint values, setting reduction targets, and tracking progress. Separating these responsibilities into layers makes each part easier to develop, test and maintain.

Reason 2 – Direct Fit for the Requirement Flow

The normal flow is structured: the resident enters consumption data, the system processes it and calculates CO₂e, monitoring components retrieve historical information, and target management uses stored values to track reduction progress. A layered design supports this sequential flow with clear interfaces between components.

Security Advantage

Authentication and authorization can be enforced before protected household data is accessed. The Resident Interface communicates through application components rather than directly accessing the Carbon Data Store, helping prevent unauthorized direct data access.

Performance Benefit

The calculation and monitoring responsibilities are separated from the presentation layer, allowing CO₂e calculations and historical-data retrieval to be handled by dedicated business components. This supports responsive interactive views and the NFR requiring annual milestones and historical graphs to render quickly.

Main Components

1. Resident Interface
2. Consumption Manager
3. CO₂e Calculator
4. Footprint Monitor
5. Target & Progress Manager
6. Carbon Data Store